

Term Project Proposal

Submit via Ninova

1. Group Information

Group Name	Databees
Member 1	Bahadır Karadağ 150240326 karadagb24@itu.edu.tr
Member 2	Yusuf Alper İLHAN 150230318 ilhan23@itu.edu.tr
Member 3	<i>Full Name Student No ITU E-mail (leave blank if group of 2)</i>

2. Dataset

Dataset Name	Student Productivity & Behavior Dataset (20K)
Source / URL	https://www.kaggle.com/datasets/algozee/student-productivity-and-behavior-dataset-20k
License	Apache 2.0
Size	20000 x 18, 1.5 MB

Dataset Description

The Student Productivity & Behavior Dataset (20K) contains 20,000 structured student records that capture information related to study habits, digital usage patterns, wellness indicators, productivity metrics, and academic results. The dataset includes variables such as daily study hours, smartphone and social media usage, sleep duration, stress levels, focus scores, productivity scores, attendance, and final grades. The dataset is synthetically generated in Python (NumPy & Pandas) to model student behavior and academic trends. This dataset is ideal for visualizing cause-and-effect relationships and behavioral patterns. It enables the use of various techniques like scatter plots, box plots, and correlation matrices to turn complex student habits into clear visual insights.

3. Motivation & Research Questions

This dataset offers a comprehensive look at a student's social and academic life. Containing 20,000 records, it provides a sufficient amount of data to capture meaningful patterns. The fact that a large portion of the dataset consists of numerical data greatly facilitates the visualization process.

The analysis will examine variables reflecting students' lifestyles such as smartphone use, social media use, and stress levels, as well as data expected to directly impact academic performance such as daily study hours and study

consistency. By analyzing the relationships between these data points the key determinants of academic success will be explored. Furthermore, students will be grouped according to specific data points revealing hidden relationships between them in detail.

Research Questions:

- 1. How does the distribution of screen time (social media, gaming) correlate with final grades?
- 2. Is there a detectable threshold where increased study hours no longer yield higher focus scores due to high stress or lack of sleep?
- 3. What are the distinct behavioral profiles of top-performing students compared to underperforming ones?

4. Planned Visualizations

Your dashboard must include at least 3 of the following visual types. Check all that apply and describe your plan for each selected type.

✓	Visual Type	Brief Description of Your Plan
☐	Time Series Plot	<i>Describe your visualization idea here ...</i>
☐	Geospatial Figure	<i>Describe your visualization idea here ...</i>
☐	Graph Visualization	<i>Describe your visualization idea here ...</i>
✓	Distribution Plot	Analyze the distribution of dataset attributes to identify the general trends among 20,000 students.
✓	Hierarchical Data	Groups student records by performance and nests behavioral habits to reveal how specific lifestyle layers and student segments contribute to academic success.
✓	Scatter Plot	Correlating digital, study, and cognitive factors with final grades to identify key performance drivers.
✓	Heatmap	Use a correlation matrix to visualize the relationships between all numerical variables

5. Tools & Technologies

Programming Language(s)	Python
Visualization Libraries	Matplotlib, Seaborn, Plotly
Dashboard Framework	Streamlit
Data Processing Tools	Pandas, NumPy

Only one group member must submit this form on Ninova.